

A novel Static Mixer for photochemical reactions

Jody Albertazzi^a, Federico Florit^b, Valentina Busini^{a,*}, Renato Rota^a

^a Politecnico di Milano, Department of Chemistry, Materials and Chemical Engineering "G. Natta", Piazza Leonardo da Vinci 32, 20133, Milano, Italy

^b Massachusetts Institute of Technology, Department of Chemical Engineering, 77 Massachusetts Avenue, Cambridge, USA

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ABSTRACT

Photochemical reactions have become a widely studied topic in the past decade, as they are employed in a wide range of processes and are recognized to be a fundamental tool in the field of sustainable chemistry. Moreover, in recent years there is an increasing interest in turning discontinuous photochemical processes into continuous ones, since continuous processes can ensure better results in terms of productivity and selectivity. In this work, a novel static mixer useful to reach a plug-flow behaviour inside a square-based reactor, while not obstructing the photon flux coming from the light sources, was designed. Through RANS based CFD simulations, the residence time distribution of molecules inside the reactor was evaluated. Moreover, the effect of both absorption and scattering phenomena, which play a major role in photochemical reactions, was assessed. Finally, a case study was considered and the photochemical degradation of 2-hydroxybenzoic acid (2-HBA) was simulated. The conversion obtained in the reactor equipped with SMs was compared to an empty reactor, finding that the degradation efficiency is doubled.

1. Introduction

Photochemical reactions are employed in a variety of industrial applications, from the production of fine chemicals (pharmaceutical, food processing) to the manufacture of basic or final products (e.g., food, electronic, packaging) [1]. Moreover, a field of growing interest is the photochemical removal of organic contaminants from water and wastewater [2–4]. Photochemistry can be also considered a key instrument to increase the sustainability of chemical processes: the synthesis of complex molecules requiring numerous steps can be significantly simplified, since photo-activated processes can obtain high complexity from simple precursors, differently from conventional thermal activated processes [2]. Many of these reactions are currently performed in discontinuous (batch or semi-batch) reactors [5], as in a batch reactor the chemical species are completely mixed, ensuring a homogeneous exposure to the photon flux. Nevertheless, continuous-flow technologies can offer several advantages with respect to discontinuous processes and their implementation is crucial for sustainable chemistry [6].

In recent years there is a tendency to shift from discontinuous processes into continuous ones [7–10], since continuous processes need reduced volumes, can ensure increased productivity, selectivity, can guarantee constant quality of the final product [11,12] and can ensure high transport phenomena [13]. In particular, among continuous reactors, a plug-flow behaviour is often preferable, since plug-flow reactors (PFRs) possess superior surface-to-volume ratio with respect to tank reactors and can be operated in harsher conditions, ensuring at the same time a higher productivity.

In order to achieve a complete plug-flow behaviour inside a chemical reactor, the hypotheses of complete radial mixing of chemical species and no axial dispersion must be simultaneously satisfied [14]. However, in many cases high residence times are required and conditions far from plug flow must be considered inside empty tubes.

In these cases, devices such as the active mixers can effectively satisfy the plug-flow conditions, but they are expensive in their operation and tend to have a high failure rate [15]. On the other hand, motionless mixers (or static mixers) are devices through which a plug-flow behaviour can be effectively obtained inside the reactor, as they can mix the reactants while, at the same time, lowering the axial dispersion: moreover, having no moving parts, these devices require low maintenance costs and need only the mechanical energy coming from the flow momentum itself [16,17]. However, even if it has been demonstrated that a tubular reactor equipped with static mixers can effectively reach a plug-flow behaviour in a relatively short distance, even considering very low values of Re [18,19], these devices are seldom used in photochemical reactions because of light obstruction [20].

Indeed in photochemical reactors, the distribution of light inside the reactor is a topic of paramount importance: whenever absorption and scattering phenomena are present (i.e., whenever the optical thickness increases), light is attenuated in the radial direction of the reactor, with strong consequences on productivity and photonic efficiency [1]. Usually, continuous photochemical reactors are designed to have a low optical thickness, defined as $thk = (k + \sigma) \cdot L$ where k and σ are the

* Corresponding author.

E-mail address: valentina.busini@polimi.it (V. Busini).



Fig. 1. Side view of the reactor.

Table 1
Geometrical properties of reactor and SM [cm].

D_{ax}	L_{ax}	D_{lat}	H_{sm1}	H_{sm2}	H_{sm3}	L_{sm1}	L_{sm2}	L_{sm3}	t_{sm}
5.08	98	1.27	3.556	1.778	5.08	5.08	3.556	1.778	0.508

absorption and scattering coefficient of the species, and L is the optical path of the reactor, respectively.

This means that continuous photochemical reactors are usually microreactors, in which the diameters are narrow, allowing the optical path to be reduced [6,21,22]. A scale-up to continuous plug-flow mesoreactors is usually difficult to be achieved and, to the best of authors' knowledge, no comprehensive studies on the possibility of obtaining a plug-flow reactor also when high optical thickness is involved (that is, roughly $thk > 3$) [3] have been presented in the literature.

The main goal of this work is to develop, through computational fluid dynamics (CFD) simulations, a novel static mixer, which can effectively attain a plug-flow behaviour in a mesoreactor while, at the same time, not hindering the photon flux from reaching the chemical species inside the reactor. A series of simulations were performed to assess the mixing length and the Residence Time Distribution (RTD) of the reactor in a various range of mass flow rates and viscosity values. Subsequently, the effect of absorption and scattering were evaluated and the performance of the reactor (in terms of conversion) equipped with the SMs was compared with an empty reactor, considering equal residence times. Finally, the same comparison was made considering as a case study the photochemical degradation of 2-hydroxybenzoic acid (2-HBA) in a slurry photocatalytic reactor.

2. Methods

The modelled geometry consists of a square-based chemical reactor equipped with sixteen static mixers and with an axial injection, as reported in Fig. 1.

Each static mixer is composed of six baffles, whose dimensions are related to the reactor diameter, as shown in Fig. 2. The first baffle has a length equal to $L_{sm1} = D_{ax}$ and a height equal to $H_{sm1} = 0.70 \cdot D_{ax}$, so that the 70% of the reactor section is occupied. The subsequent two elements are placed so that the flow is forced to be twisted downwards: each element has a length equal to $L_{sm2} = D_{ax}$ and a height equal to $H_{sm2} = 0.35 \cdot D_{ax}$, so that the section occupied by the sum of these two baffles is the 70% of the reactor section. The subsequent baffles are placed in order to twist the flow left and right: the fourth baffle has a height equal to $H_{sm3} = D_{ax}$ and a length equal to $L_{sm2} = 0.70 \cdot D_{ax}$, while the last two baffles have a height equal to H_{sm3} and a length equal to $L_{sm3} = 0.35 \cdot D_{ax}$. The aspect ratio of the SM is equal to 1, meaning that the distance between the first baffle of each SM is equal to D_{ax} , while the thickness of each baffle is equal to $t_{sm} = D_{ax}/10$. Table 1 resumes the geometrical properties of both the reactor and the SMs.

The streams labelled A and B in Fig. 1 have diverse properties in terms of mass flow rates and viscosity. This way, various values of Re are obtained, defined as:

$$Re = \frac{\rho_{mix} \cdot v \cdot D_{ax}}{\mu_{mix}} \quad (1)$$

where the mixture density and the mixture viscosity are obtained through mass weighted averages, given respectively by $\rho_{mix} = (\omega_A/\rho_A +$

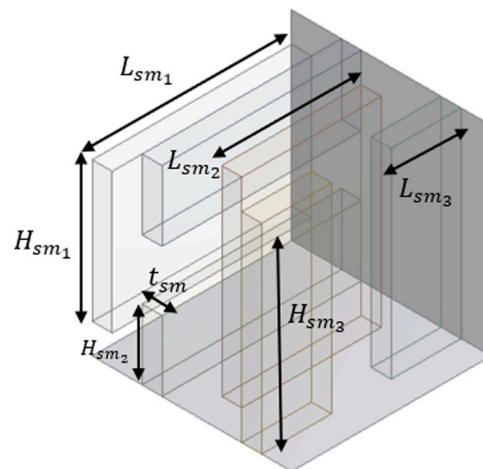


Fig. 2. Static mixer geometry.

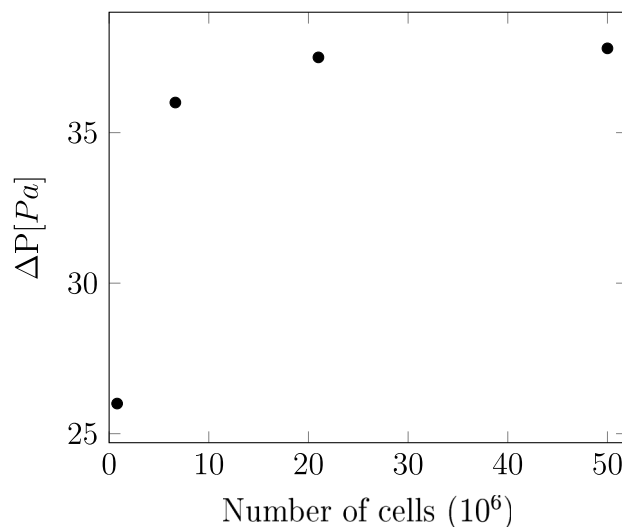


Fig. 3. Pressure drop as function of number of cells.

$\omega_B/\rho_B)^{-1}$ and $\mu_{mix} = \mu_A \omega_A + \mu_B \omega_B$, ω_A and ω_B being the mass fraction of the chemical species A and B, given by $\omega_A = \dot{m}_A/(\dot{m}_A + \dot{m}_B)$ and $\omega_B = \dot{m}_B/(\dot{m}_A + \dot{m}_B)$.

2.1. Mixing length and residence time distribution analysis

The mixing length was evaluated through the coefficient of variation COV, evaluated as:

$$COV(x) = \frac{\sqrt{(\omega_{B,x}^2 - \overline{\omega_{B,x}}^2)}}{\overline{\omega_{B,x}}} \quad (2)$$

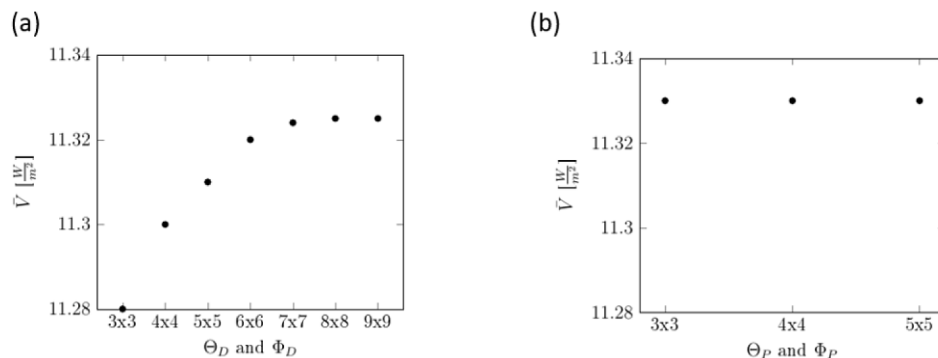


Fig. 4. Volume weighted incident radiation as a function of Θ_D and Φ_D (a) and Θ_P and Φ_P (b).

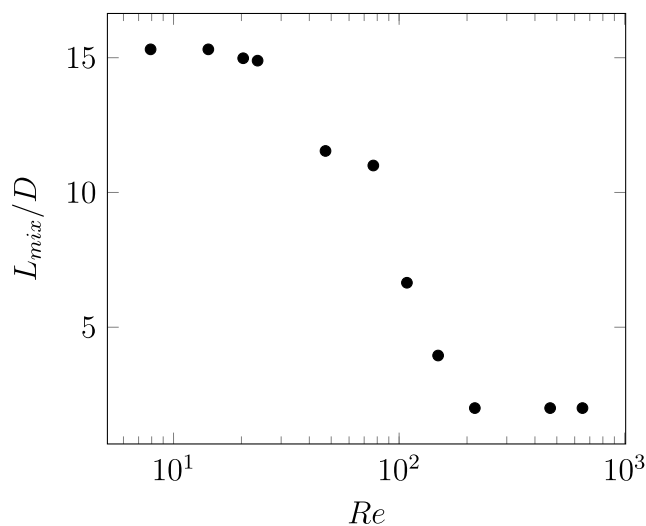


Fig. 5. L_{mix} as a function of Re .

where $\overline{\omega_{B,x}^2}$ is the mass-flow average (on a cross section of the reactor) of the squared mass fraction of B calculated at the axial position x and $\overline{\omega_{B,x}}$ the mass-flow average (on the same section) of the mass fraction of B calculated at the axial position x . Through the measure of this coefficient, it is possible to evaluate the length needed to mix almost completely the chemical species inside the reactor [23], since when $COV \leq 0.05$ complete mixing can be assumed, while for high values of COV the flow is segregated. In this work, a non dimensional value of the mixing length was considered and the axial position at which $COV \leq 0.05$ was divided by the characteristic size of the static mixer, D_{ax} , in order to obtain the numbers of static mixers necessary to achieve complete mixing, $N = \frac{L_{mix}}{D_{ax}}$. Finally, as done in previous works [18,19], also the residence time distribution was assessed for the reactor equipped with static mixers. The results were also compared with those of the mixing length analysis, through the simulation of an unsteady-state tracer injection.

2.2. CFD simulations

2.2.1. Hydrodynamics and mass transfer

In this work, the Ansys Fluent 19.1 CFD software was used to solve the continuity, momentum, and species mass fraction transport equations through RANS-based simulations. Since the static mixers are capable to induce turbulence also for laminar values of Re , a closure equation for the time-averaged equations is needed: Shear-Stress Transport (SST) $k-\omega$ model is known to be able to model fluids within a wide range of fluid dynamic regimes [24] and it was chosen for this work.

Table 2

Operating conditions and fluid properties for grid-independence test.

$\rho_A \left[\frac{kg}{m^3} \right]$	$\rho_B \left[\frac{kg}{m^3} \right]$	$\mu_A \text{ [cP]}$	$\mu_B \text{ [cP]}$	$\dot{m}_A \left[\frac{kg}{s} \right]$	$\dot{m}_B \left[\frac{kg}{s} \right]$
1000	1000	0.01	0.001	0.01	0.001

All the other solver parameters (e.g., turbulence model parameters, under-relaxation factors, and internal discretization parameters) were left to the default values. The boundary conditions for the walls of the reactor and the SMs are non-slip walls with null species fluxes. The inlet boundary conditions (both axial and lateral) are set as mass flow inlet with assigned composition, while the outlet is set as a pressure outlet with null gauge pressure and backflow composition equal to the composition obtained when the axial and lateral streams are perfectly mixed. Convergence of the steady-state simulations (used in the mixing length analysis) was checked by visually observing low and constant residuals of the transport equations and by verifying the overall mass balance (i.e., mass conservation through the fluxes entering and exiting the reactor). Transient simulations (used for the RTD determination) were performed using a constant time-stepping satisfying the Courant–Friedrichs–Lewy condition, that is, $Co = u \cdot \Delta t / h < 1$ [25], where Co is the Courant number, Δt the integration time step and h the cell size; through a time-stepping-independence analysis, a time step of 0.03 s was selected and 25 iterations were allowed for each time step.

The unstructured grid (mesh) is composed of tetrahedral cells: in order to have reliable results which are independent of the grid, a mesh independence test, needed to determine the minimum number of cells (or the minimum dimension of cells), needs to be performed. The test was performed by measuring the pressure drop at the exit of the reactor, since pressure drops have been reported to be particularly susceptible to perturbations [26]. Grids having 0.08, 5.5, 21 and 50 million cells were considered for this test. The boundaries conditions adopted are reported in Table 2.

The obtained results, reported in Fig. 3, show that over 20 million cells the change in the results is negligible, which means that this number of cells is the minimum to get results independent of the grid size.

2.2.2. Radiation modelling

The CFD modelling of photoreactors also needs the evaluation of the radiation field inside the reactors: this is possible through the solution of the radiative transfer equation (RTE):

$$\frac{dI(\vec{r}, \vec{s})}{ds} + (k + \sigma_s)I(\vec{r}, \vec{s}) = kn^2 \frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi} \int_0^{4\pi} I(\vec{r}, \vec{s}') \Phi(\vec{s} \cdot \vec{s}') d\Omega' \quad (3)$$

where $\vec{s}$ and $\vec{r}$ are the position vectors, $\vec{s}'$ is the scattering direction vector, s is the path length, n is the refractive index, σ_s is the scattering coefficient, k is the absorption coefficient, σ is the Stefan–Boltzmann constant, I is the radiation intensity, which depends on position vectors

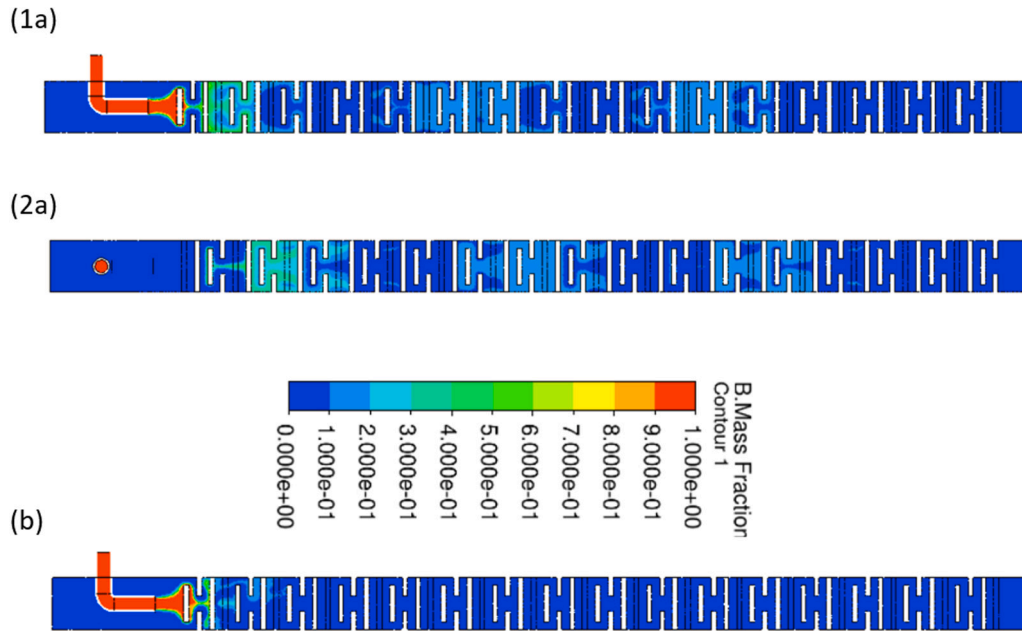


Fig. 6. B mass fraction contour for: $Re = 14$ in the xy plane (1a), $Re = 14$ in the xz plane (2a) and $Re = 216$ (b).

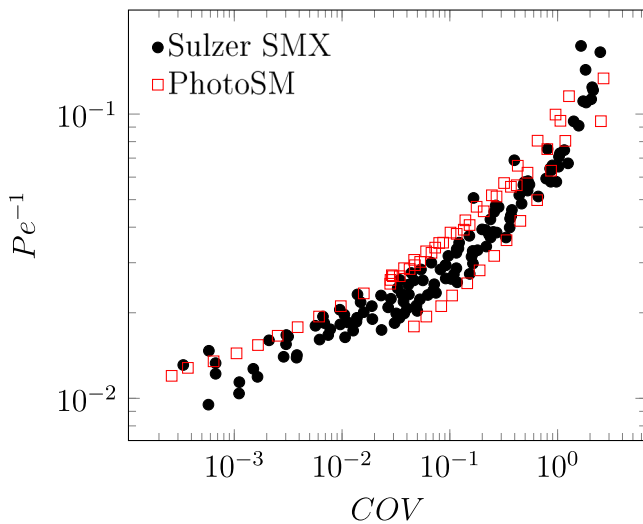


Fig. 7. Pe^{-1} as a function COV comparing the Sulzer SMX with the proposed configuration (PhotoSM).

$\vec{r}$ and $\vec{s}$, T is the local temperature, Φ the phase function and Ω' the solid angle [24]. There are various models useful to solve the RTE: among them, the Discrete Ordinate Model (DOM) can give an accurate solution of the RTE [27] and it was chosen for this work.

This method solves the RTE dividing by a finite number of discrete solid angles, each associated with a vector direction fixed in the global Cartesian system $(\vec{x}, \vec{y}, \vec{z})$ [24]. It is necessary to use an angular discretization to avoid the ‘ray-effect’ and a pixelization high enough to avoid the angle overhanging [28]: in order to have reliable results, an independence test was performed. Two walls of the reactor shown in Fig. 1 were set as semi-transparent with purely diffusive irradiation of $I = 60 \text{ W/m}^2$, while all the other walls were set as semi-transparent with a diffusion coefficient equal to 1; the fluid inside the reactor was set with a coefficient of absorption equal to $k = 200 \text{ m}^{-1}$ and a scattering coefficient equal to $\sigma = 50 \text{ m}^{-1}$. Since the semi-transparent walls and the mixer walls were modelled as purely diffusive, the light entering the

Table 3

Operating conditions and fluid properties for the mixing-length analysis.

$\dot{m}_A$ [kg/s]	μ_A [Pa s]	$\dot{m}_B$ [kg/s]	μ_B [Pa s]
$1 \cdot 10^{-2}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
$1 \cdot 10^{-3}$	$2 \cdot 10^{-3}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
$1 \cdot 10^{-2}$	$3 \cdot 10^{-2}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
$5 \cdot 10^{-3}$	$8 \cdot 10^{-3}$	$5 \cdot 10^{-4}$	$1 \cdot 10^{-3}$
$1 \cdot 10^{-2}$	$2 \cdot 10^{-2}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
$5 \cdot 10^{-3}$	$1 \cdot 10^{-3}$	$5 \cdot 10^{-4}$	$1 \cdot 10^{-3}$
$1 \cdot 10^{-1}$	$5 \cdot 10^{-3}$	$1 \cdot 10^{-2}$	$1 \cdot 10^{-3}$
$2 \cdot 10^{-2}$	$1 \cdot 10^{-2}$	$2 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
$3 \cdot 10^{-2}$	$1 \cdot 10^{-3}$	$3 \cdot 10^{-3}$	$1 \cdot 10^{-3}$
$5 \cdot 10^{-3}$	$7 \cdot 10^{-4}$	$5 \cdot 10^{-4}$	$1 \cdot 10^{-3}$
$1 \cdot 10^{-2}$	$3 \cdot 10^{-3}$	$1 \cdot 10^{-3}$	$1 \cdot 10^{-3}$

reactor and the light reflected by the mixing elements is not directed along a preferential axis, but is scattered in all directions. The static mixer can be considered as having the properties of a reflective metal, such as polished aluminium. Moreover, to inactivate calculations of radiation emission the temperature was fixed to 1 K in all domains [3, 28], and the volume-weighted incident radiation ($\bar{V}$) was monitored. First, the pixelization value in the azimuthal direction and in the radial direction (θ_p and ϕ_p , respectively) were kept constant and with a value of 3×3 , while the discretization in the azimuthal direction and in the radial direction (θ_D and ϕ_D , respectively) were varied. Subsequently, the values of θ_p and ϕ_p were varied while keeping constant θ_D and ϕ_D . Fig. 4a shows that using an angular discretization of the sphere octant of 8×8 allows to obtain negligible changes in the volume-weighted incident radiation: moreover, with these values of θ_D and ϕ_D the results were found to be independent of the pixelization in the radial and azimuthal direction (Fig. 4b).

3. Results and discussion

3.1. Mixing

A mixing length analysis was carried out to assess whether this static mixer can effectively mix the streams also when fluids with relatively different properties, in terms of mass flow rates and viscosity, are considered. The boundary conditions adopted are summarized in Table 3.

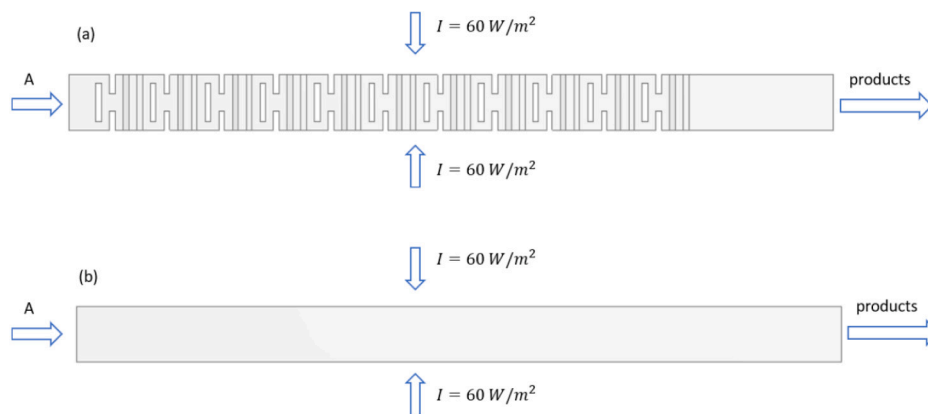


Fig. 8. Side view of the tubular reactor with SMs (a) and without SMs (b).

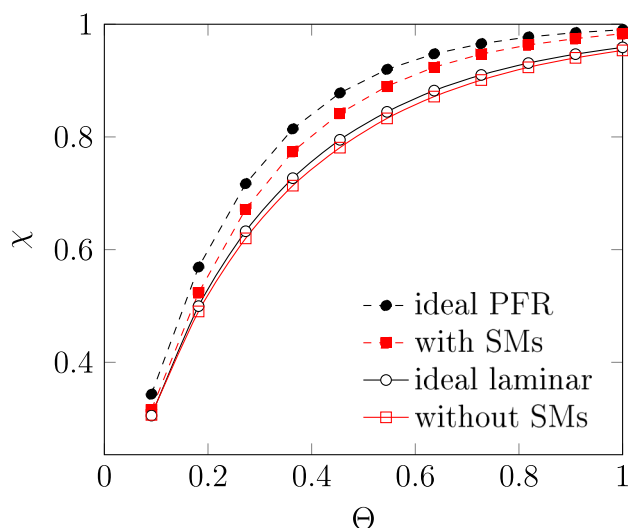


Fig. 9. Conversion profiles of ideal PFR and ideal laminar reactor compared to conversion profiles obtained in reactor equipped with SMs and in the reactor without SMs.

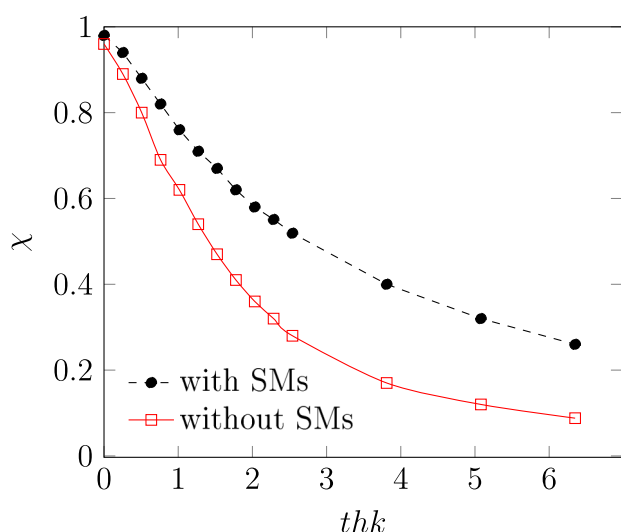


Fig. 10. Conversion values vs various optical thicknesses.

Results are reported in Fig. 5, which shows the values of L_{mix} as a function of Re . The mixing lengths are comprised in a range of lengths

of 2–15 static mixers and two plateaus are reached for, roughly, $Re < 30$ and $Re > 200$. We can see that after a maximum of 15 static mixers the hypotheses of complete mixing can be considered fulfilled, even for very low values of Re .

The trend obtained is coherent with the definition of Re , which can characterize the fluid dynamics in the static mixers [29]: when the inertial forces are predominant over the viscous forces, the mixing process will be favoured, as shown in Fig. 6. This trend was also found in other static mixers [18,19].

3.2. Residence time distribution

A RTD analysis was performed, similarly to what discussed in previous works [18,19]. The obtained distributions were interpreted using the tanks-in-series model (TISM) and the axial-dispersion model (ADM) [14]. The results are reported in Table 4, in terms of both the number of equivalent $CSTRs$, which is defined as the number of $CSTRs$ that need to be put in series in order to approximate the reactor performance, and the inverse of the Peclet number, which is defined as the ratio between the advective transport rate (given by the product between the characteristic length L (in our case, the reactor diameter D) and the spatial velocity v) and the diffusion transport rate (given by the axial diffusion coefficient $\mathcal{D}_{ax}$): $Pe = Lv/\mathcal{D}_{ax}$. We can see that even for creeping flows ($Re < 10$), which is a condition that does not favour a plug flow behaviour inside the reactor, a high value of N_{CSTRs} and a low value of Pe^{-1} is obtained.

Moreover, at increasing Re , the axial dispersion inside the reactor (i.e., Pe^{-1}) decreases and the number of equivalent mixed vessels (i.e., N_{CSTRs}) increases, coherently to what reported in previous studies for different Static Mixers [18,19]. Another significant information is the fraction of dead volume, which represents the number of stagnant zones inside the reactor and can be calculated as:

$$V_{dead} = \frac{(\tau - \bar{t}) \cdot Q}{V_{reactor}} \quad (4)$$

where τ is the ratio of the reactor volume to the total volumetric flowrate, Q the volumetric flowrate, and $V_{reactor}$ the reactor volume. As reported in Table 4, the value of V_{dead} is negligible also when a strongly viscous fluid is considered.

Notably, COV reached a value equal to 0.05 when the value of Pe^{-1} was roughly equal to 0.02, in all the cases considered. This suggests that a correlation between the radial dispersion (measured by COV) and the axial dispersion (measured by Pe^{-1}) could exist, as shown in Fig. 7.

It is possible to see that the values of Pe^{-1} grow monotonically at increasing COV : both values of COV and Pe^{-1} decrease as the flow proceeds through the static mixers. When the chemical species are completely mixed (that is, $COV \leq 0.05$), the axial dispersion can be assumed negligible as well. Notably, the same trend was obtained in a previous work in which a series of fluid dynamics simulations were performed on a side-injection tubular reactor equipped with SMX [18,19], as shown in the same Fig. 7.

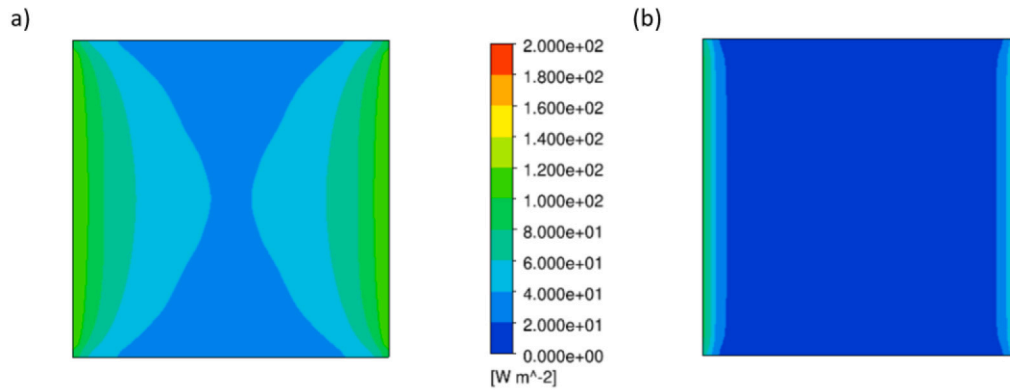


Fig. 11. Incident radiation contour on the cross section of the reactor for $k_{mix} = 30 \text{ m}^{-1}$ (a) and $k_{mix} = 250 \text{ m}^{-1}$.

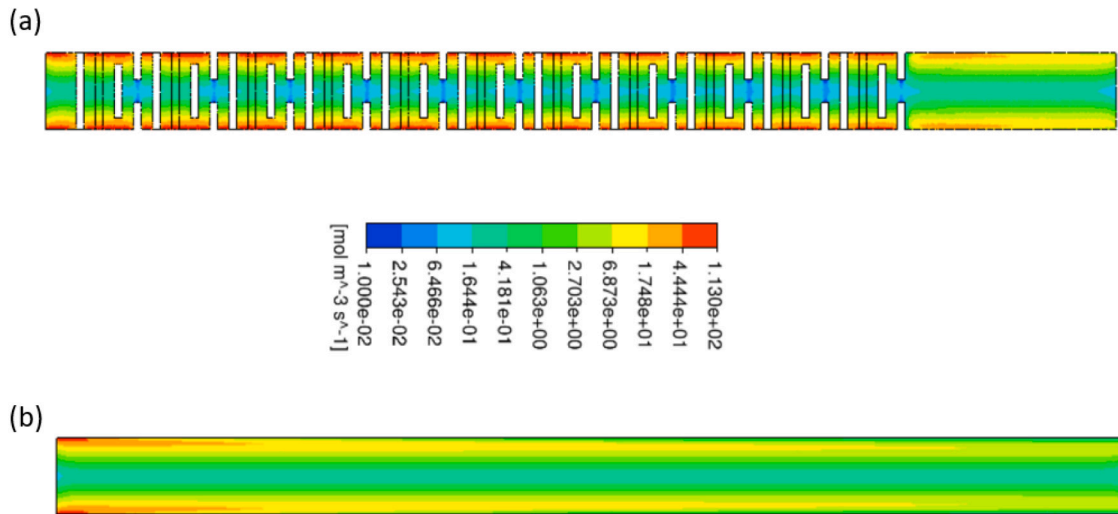


Fig. 12. Contour in logarithmic scale of the rate of reaction for the reactor with SMs (a) and without SMs (b).

Table 4

Operating conditions and results for the RTD analysis.

$\dot{m}_A$ [kg/(m s)]	$\dot{m}_B$ [kg/(m s)]	μ_A [cP]	μ_B [cP]	Re	Pe^{-1}	N_{CSTRs}	$V_{dead}\%$
$1.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-1}$	$1.0 \cdot 10^{-3}$	2	$2.4 \cdot 10^{-2}$	22	0.2
$1.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	$5.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	5	$2.8 \cdot 10^{-2}$	22	0.3
$1.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	23	$2.2 \cdot 10^{-2}$	27	0.2
$1.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	$3.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	76	$1.7 \cdot 10^{-2}$	32	0
$1.0 \cdot 10^{-2}$	$1.0 \cdot 10^{-3}$	$2.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	113	$2.9 \cdot 10^{-2}$	21	0.3

3.3. Photochemical reaction

In order to model the behaviour of a photochemical reactor, where a heterogeneous field of the reaction rate may exist, absorption and scattering phenomena that can lead to an attenuation of light inside the reactor must be accounted for [1].

Therefore, to assess if the presence of the static mixer can hinder the light distribution inside the reactor, the conversion obtained in a tubular reactor equipped with SMs (Fig. 8A) were compared to the ones obtained within a tubular reactor without SMs (Fig. 8B) at equal values of residence time.

The four sides of the reactor were simulated as semi-transparent; incoming radiation equal to $I = 60 \text{ W/m}^2$ was imposed on two of the four sides of the reactor. The static mixers were simulated as opaque walls with a diffuse fraction equal to 1. The generic reaction considered in this analysis together with its reaction rate is:

$$r = K \cdot C_A \cdot LVRPA \quad (5)$$

where K is the kinetic constant, equal to $K = 4 \cdot 10^{-5} \text{ m}^3/\text{W/s}$, C_A kmol/m³ is the concentration of chemical species A , and $LVRPA$ is the Local Volumetric Rate of Photon Absorption, given by the integral of the radiation intensity over the solid angle multiplied by the absorption coefficient:

$$LVRPA = k_A \int_0^{4\pi} I d\Omega \quad (6)$$

where k_A is the absorption coefficient of the A component, which has been assumed constant and equal to $k_A = 1 \text{ m}^{-1}$. The reactor was simulated implementing a suitable user-defined function, UDF , in Fluent.

3.4. Light obstruction

To verify if the presence of the SMs causes significant hindrance to the light penetration, a simulation was run imposing the mixture absorption $k_{mix} = 0 \text{ m}^{-1}$ and the mixture scattering $\sigma_{mix} = 0 \text{ m}^{-1}$. In this case, since the absorption is negligible inside the reactor, the mean

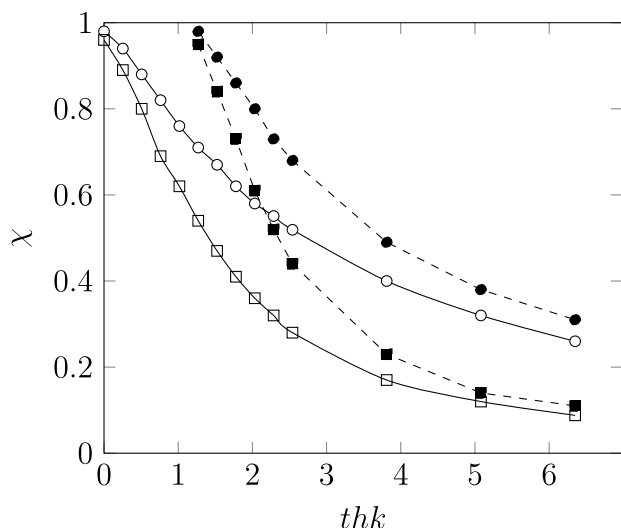


Fig. 13. (●): conversion values with $\sigma_{mix} = 50 \text{ m}^{-1}$ in reactor with SMs; (■): conversion values with $\sigma_{mix} = 50 \text{ m}^{-1}$ in reactor without SMs; (○): conversion values with $\sigma_{mix} = 0 \text{ m}^{-1}$ in reactor with SMs; (□): conversion values with $\sigma_{mix} = 0 \text{ m}^{-1}$ in reactor without SMs.

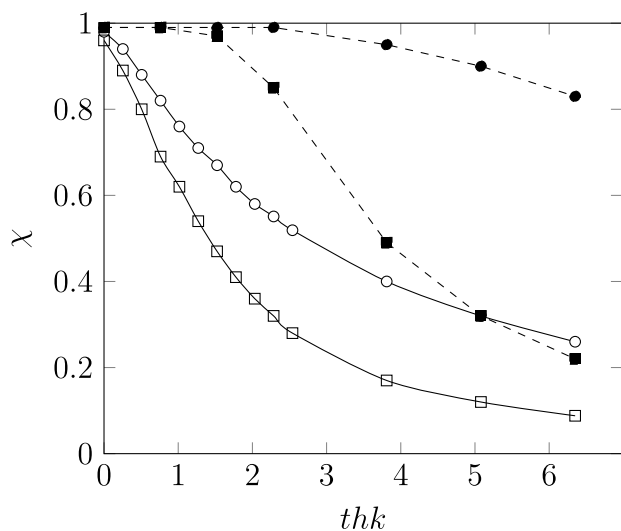


Fig. 14. conversion vs thickness for $\sigma_{mix} = 0$: (●): conversion values with K_{10} in reactor with SMs; (■): conversion values with K_{10} in reactor without SMs; (○): conversion values with K in reactor equipped with SMs; (□): conversion values with K in reactor without SMs.

incident radiation will be equal to $\bar{I} = 120 \text{ W/m}^2$. A simulation was performed and the conversion values were evaluated along the longitudinal direction of both the reactor equipped with SMs and the reactor without SMs. In this simulation, $\dot{m}_A = 0.01 \text{ kg/s}$, $\rho_A = 1000 \text{ kg/m}^3$, $\mu_A = 0.001 \text{ Pas}$ and $k_{mix} = \sigma_{mix} = 0 \text{ m}^{-1}$.

Knowing that, in an ideal PFR with a first-order reaction, the obtained conversion can be expressed as $X_{PFR} = 1 - \exp(-K \cdot k_A \cdot \bar{I} \cdot \tau)$ and in an ideal laminar reactor the conversion is equal to $X_{lam} = 1 - y^2 \cdot \text{ei}(y) + (1 - y) \cdot \exp(-y)$, where $y = (\frac{K \cdot k_A \cdot \bar{I} \cdot \tau}{2})$ and $\text{ei}(y)$ is the exponential function [14], it is possible to plot the conversion along the reactor and to compare the theoretical values with the ones obtained through the simulations. The non-dimensional residence time $\theta = t/\tau$, where $\tau = V/Q$ (being V the reactor volume and Q the volumetric flow rate) was considered.

The results are reported in Fig. 9: we can see that, as expected, the profile of conversion obtained in the reactor equipped with SMs

and in the reactor without SMs are close to the PFR ideal model and the laminar model, respectively. Therefore, it is possible to conclude that the SMs do not hinder significantly the light distribution inside the reactor.

3.5. Absorption

To investigate the effect of absorption phenomena over the conversion the absorption coefficient of the mixture k_{mix} was set with increasing values and the conversion after the last static mixer was compared with the conversion obtained in the reactor without SMs for equal residence times. In this set of simulations, $\dot{m}_A = 0.01 \text{ kg/s}$, $\rho_A = 1000 \text{ kg/m}^3$, $\mu_A = 0.001 \text{ Pas}$ and $\sigma_{mix} = 0 \text{ m}^{-1}$, while k_{mix} was varied between 0 and 250 m^{-1} .

The results are reported in Fig. 10, in which the conversions are plotted against thk , which is defined as $thk = (k_{mix} + \sigma_{mix}) \cdot D/2$. As expected, it is clear how the increase of the absorption coefficient leads to lower conversion since the light is exponentially attenuated inside the reactor, as shown in Fig. 11. However, the presence of SMs leads to recirculation of the reactant to the highly irradiated sides of the reactor: this means that, while in the reactor without SMs the reaction rate diminishes along the longitudinal direction of the reactor, the effect of SMs can ensure a more homogeneous profile of reaction rate along the axial coordinate and thus a higher conversion, as shown in Fig. 12. The reaction in the reactor without SMs is limited to reactants close to the wall; the reaction almost vanishes after a few diameters as the radiation does not penetrate inside the reactor and no fresh reactant is supplied to the walls.

3.6. Scattering

To investigate the effect of the scattering, the previous simulations carried out at different values of k_{mix} and $\sigma_{mix} = 0$ were repeated by setting $\sigma_{mix} = 50 \text{ m}^{-1}$. In the whole set of simulations, $\dot{m}_A = 0.01 \text{ kg/s}$, $\rho_A = 1000 \text{ kg/m}^3$, $\mu_A = 0.001 \text{ Pas}$, k_{mix} was varied between 0 and 200 m^{-1} and σ_{mix} (which is considered as isotropic) was kept constant and equal to 50 m^{-1} .

The results are reported in Fig. 13: we can see that, for equal values of thk , the scattering has a beneficial effect on the conversion. This is consistent with the form of the RTE (Eq. (3)): indeed, in this case, the mean incident radiation inside the reactor is higher, given the in-scattering phenomena that take place.

3.7. Chemical kinetics

To investigate the effect of the chemical kinetics, a set of simulations, with a higher kinetic constant (in particular, the new kinetic constant $K_{10} = K \cdot 10$) has been carried out by changing the k_{mix} values and considering $\sigma_{mix} = 0$ and $\sigma_{mix} = 50 \text{ m}^{-1}$, as with previous simulations. In the whole set of simulations, $\dot{m}_A = 0.01 \text{ kg/s}$, $\rho_A = 1000 \text{ kg/m}^3$, $\mu_A = 0.001 \text{ Pas}$, $\sigma_{mix} = 0 \text{ m}^{-1}$ and k_{mix} was varied between 0 and 250 m^{-1} .

The results are summarized in Fig. 14 for $\sigma_{mix} = 0$: we can see that the beneficial effect of the SMs is higher when faster reactions are involved. In the reactor without SMs, as shown in Fig. 15b, the chemical species B is immediately produced at the entrance of the reactor and the reaction cannot continue, since the light diffusion is considerably hindered by the mixture absorption. On the other hand, SMs are able to recirculate the reactant close to the light source (Fig. 15a), leading to much higher conversion values.

The results for $\sigma_{mix} = 50 \text{ m}^{-1}$ are reported in Fig. 16. We can see that, as expected, the same trends found with $\sigma_{mix} = 0 \text{ m}^{-1}$ are obtained.

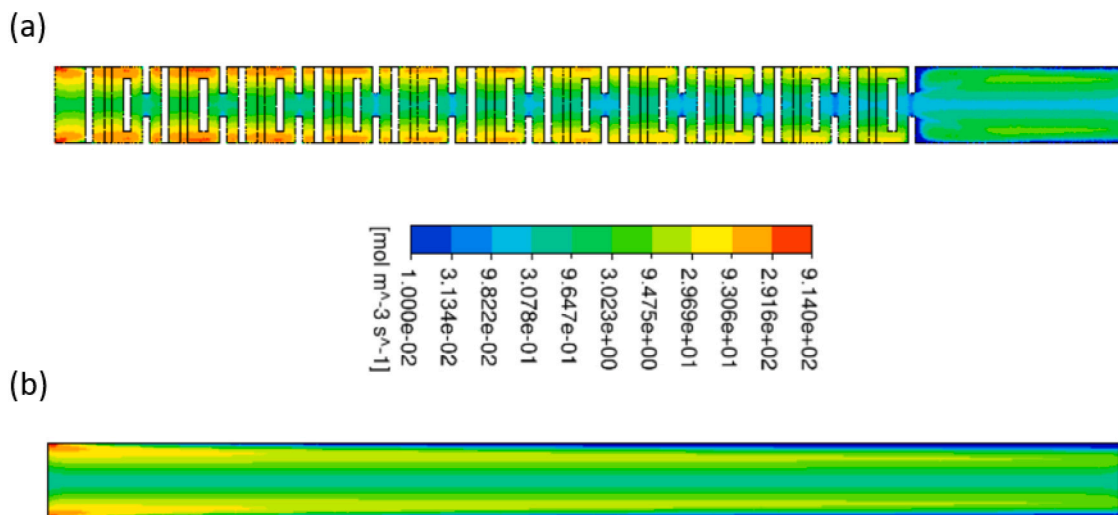


Fig. 15. Contour in logarithmic scale of kinetic rate of reaction: (a) K_{10} in the reactor with SMs; (b) K_{10} in the reactor without SMs.

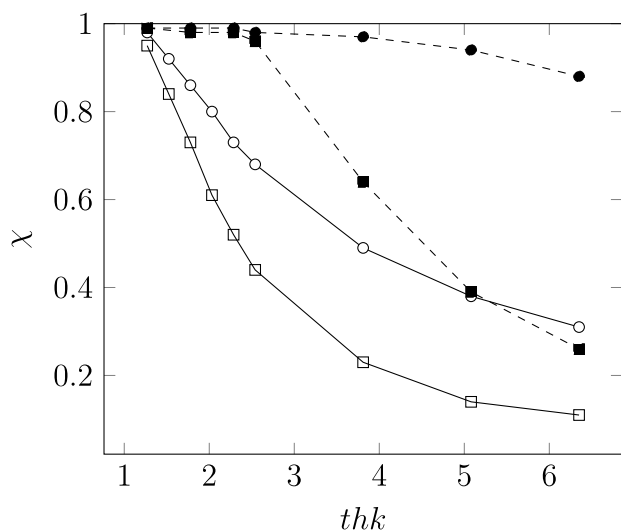


Fig. 16. conversion values vs thk for $\sigma_{mix} = 50 \text{ m}^{-1}$: (●): K_{10} in reactor with SMs; (■): K_{10} in reactor without SMs; (○): K in reactor with SMs; (□): K in reactor without SMs.

4. Case study

The photocatalytic oxidation of 2-hydroxybenzoic acid (2-HBA) has been investigated as a case study to show the effectiveness of the proposed reactor configuration. This toxic aromatic organic compound is widely used as an intermediate in the pharmaceutical and cosmetic field [30] and is found in many industrial effluents [31], polluting the environment and causing problems such as tinnitus, qualm and naupathia if brought into contact with people [30].

In particular, the photochemical oxidation of 2-HBA using a suspension of TiO_2 P25 photocatalyst was simulated. The catalyst suspension can be considered as a pseudo-homogeneous medium [28,32] and both the absorption and scattering coefficient can be considered independent of position and time [27]. A dual-wavelength band interval was considered, dividing the radiation spectrum into two parts, UVA (315 nm–400 nm) and UVB (280 nm–315 nm) [3]. Through the multiplication of the absorption coefficient for the incident radiation, it is possible to obtain the local volumetric photon absorption for the UVA band ($LV RPA_{UVA}$) and the local volumetric photon absorption for the UVB band ($LV RPA_{UVB}$). A mass flow rate equal to $m_{in} = 1 \cdot 10^{-5} \text{ kg/s}$ was considered, while the density and viscosity of the catalyst

Table 5

Operating conditions and optical properties for 2-HBA degradation.

C_{BHA} [mmol/L]	C_{cat} [kmol/m ³]	k_a^* [m ² /kg]	σ_a^* [m ² /kg]
0.2	0.1	189.9	508.5
0.2	0.2	189.9	508.5

suspension were set equal to $\rho = 998 \text{ kg/m}^3$ and $\mu = 0.001 \text{ kg/m}^3$. Coherently to the simulations previously performed, the effect of SMs was compared to that of the reactor without SMs at equal residence times.

The optical properties of the TiO_2 P25 suspension are reported in Table 5.

From specific absorption coefficient (k_a^*) and the specific scattering coefficient (σ_a^*) it is possible to obtain the Local Volumetric Rate of Photon Absorption for each wavelength ($LV RPA_\lambda$):

$$LV RPA_\lambda = k^* \cdot C_{cat} \int_0^{4\pi} I_\lambda d\Omega \quad (7)$$

The reaction rate can be computed as [3]:

$$-r_{2-HBA} = \frac{k_1 K_{L-H}}{1 + K_{L-H} C_{2-HBA}} \cdot C_{2-HBA} [(LV RPA_{UVA})^m - (LV RPA_{UVB})^m] \quad (8)$$

where k_1 [kmol m^{-1.5} s⁻¹ W^{-0.5}] is the intrinsic kinetic constant of photocatalytic oxidation of 2-HBA, K_{L-H} [m³ kmol⁻¹] is the Langmuir–Hinshelwood binding constant for TiO_2 P25, C_{2-HBA} [kmol m⁻³] is the concentration of 2-HBA and m is a non-dimensional coefficient equal to 0.5 for moderate to intense photon fluxes [31,33]. Both the reaction kinetic and the scattering coefficient were included in the CFD model through two dedicated UDFs.

The results are reported in Fig. 17: we can see that, in both cases, there is a significant difference between reactors with and without SMs. In particular, the final conversion in the reactor without SMs is about half the conversion measured in the reactor with SMs. Moreover, when the catalyst concentration is doubled (Fig. 17b) the conversion values reduce according to the trends previously reported in the literature [3, 34]. This behaviour is due to the larger radiation shuttering induced by higher catalyst concentrations.

5. Conclusions

A novel static mixer specifically designed to handle photochemical reactions involving high optical thicknesses was designed and

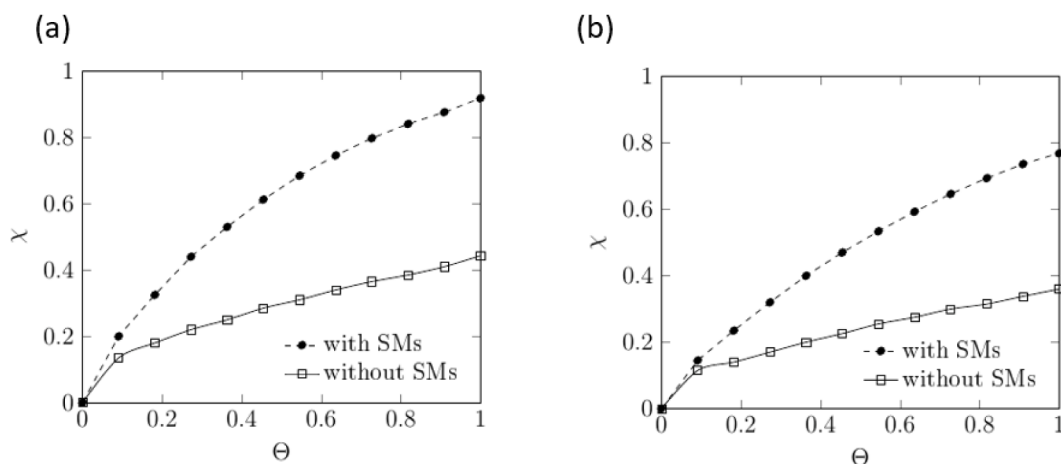


Fig. 17. Conversion values vs non dimensional residence time Θ : (a) catalyst concentration equal to 0.1 kmol/m³; (b) catalyst concentration equal to 0.2 kmol/m³.

its efficiency in terms of mixing length, *RTD*, light obstruction, and expected conversion was evaluated. It was found that with this device it is possible to obtain substantially higher conversions with respect to an empty reactor for high values of optical thickness. Indeed, since this new type of SM can effectively recirculate the reactant species to the light source, the conversion drop is much less severe when the absorption, scattering, and optical path of the reactor increase. This device could be particularly useful to transform a batch photochemical process into a continuous one, since a plug flow behaviour is reached with negligible light obstruction. Moreover, the reactor with SMs is capable to reach higher conversions for each load of catalyst considered in this work; this means that the optimum load of catalyst in the reactor with SMs (which is a critical factor in the design and optimization of a photochemical reactor, since it maximizes the conversion without hindering the light diffusion) is higher than in the reactor without SMs.

The static mixer presented in this work can be employed in the construction of a multi-layered square-based mesoreactor; in this case, two sides of the reactor should be built with an opaque material such as steel, to allow the layering of the modules. Alternatively, this type of SM can be inserted in a photocatalytic reactor operated in a flow-through recirculation batch mode [3,28], since it may considerably increase the conversion per single passage.

CRedit authorship contribution statement

Jody Albertazzi: Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Federico Florit:** Conceptualization, Formal analysis, Software. **Valentina Busini:** Conceptualization, Data curation, Supervision, Writing – review & editing. **Renato Rota:** Conceptualization, Formal analysis, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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